

Sports Action Recognition Framework Using Deep Learning

Kavana Chakkere Venkatesh
Matriculation Number: 93759429
University of Europe for Applied Sciences
Germany

Abstract—This paper presents a Sports Action Recognition Framework using deep learning techniques. A Convolutional Neural Network (CNN) is employed to recognize and classify human actions from image data. The study evaluates the effectiveness of deep learning for automated action recognition. The proposed framework was evaluated on a dataset containing 12,600 images across 15 human activity classes.

I. INTRODUCTION

Sports action recognition is an important computer vision task. Sports action recognition is widely used, in intelligent surveillance systems, sports analytics, healthcare monitoring, human-computer interaction, and video understanding applications.

Recent advances in deep learning have significantly improved the ability of machines to automatically recognize human activities from visual data. Convolutional Neural Networks (CNNs) have become one of the most effective approaches for extracting meaningful features from images and performing accurate classification.

II. DATASET

The dataset used is the Sports and Human Activity Recognition dataset from Kaggle.

III. METHODOLOGY

A Convolutional Neural Network (CNN) was employed for sports action recognition and image classification. The proposed framework consists of dataset preprocessing, image normalization, CNN model construction, model training, and performance evaluation.

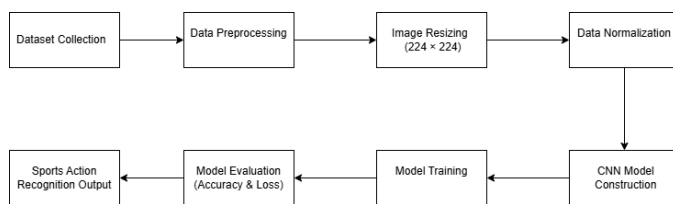


Fig. 1. Methodology of the Proposed Sports Action Recognition Framework

Figure 1 illustrates the workflow of the proposed Sports Action Recognition Framework. The process begins with dataset collection and preprocessing, followed by image resizing and normalization. A CNN model is constructed and trained using the processed images. Model performance is then evaluated

using accuracy and loss metrics before generating the final sports action recognition output.

CNNs are particularly suitable for image classification because they automatically learn spatial features through convolution operations. Pooling layers reduce dimensionality while preserving important information, and dropout regularization helps prevent overfitting.

IV. RESULTS

The initial training results obtained from the CNN model are presented and discussed in this section.

The initial training accuracy of 9.20% indicates that the model had just begun learning useful feature representations. During the early stages of training, CNN models typically exhibit lower accuracy because network weights are still being optimized. The corresponding training loss was 3.0105, reflecting the model's initial learning phase. Additional training epochs are expected to improve classification performance by reducing loss and increasing prediction accuracy. The reported results serve as a baseline for further model optimization and performance enhancement.

V. RELATED WORK

Deep learning has revolutionized image classification tasks. Convolutional Neural Networks (CNNs) have demonstrated outstanding performance in extracting visual features from images. Popular architectures such as AlexNet, VGG16, ResNet, MobileNet, and EfficientNet have been widely applied in image recognition applications.

Several studies have applied deep learning techniques to sports image classification. Traditional machine learning approaches relied on handcrafted features, while modern CNN-based approaches automatically learn hierarchical feature representations. These deep learning methods generally achieve higher accuracy and better generalization performance.

Previous studies have explored CNN-based approaches for human activity recognition using image and video datasets. Researchers have demonstrated that deep learning models outperform traditional machine learning methods because they automatically learn hierarchical feature representations.

Transfer learning approaches using architectures such as ResNet, VGG16, MobileNet, and EfficientNet have also been successfully applied to action recognition tasks. These models achieve high classification accuracy while reducing training time and computational requirements.

VI. DATASET DESCRIPTION

The dataset used is the Sports and Human Activity Recognition dataset from Kaggle. The dataset contains 12,600 images representing 15 different human activities. The images were organized into training and testing sets to facilitate supervised learning.

The dataset includes the following activity classes:

- Calling
- Clapping
- Cycling
- Dancing
- Drinking
- Eating
- Fighting
- Hugging
- Laughing
- Listening to Music
- Running
- Sitting
- Sleeping
- Texting
- Using Laptop

The dataset was split into 10,080 training images and 2,520 testing images.

A. Dataset Statistics

TABLE I
DATASET STATISTICS

Property	Value
Total Images	12,600
Training Images	10,080
Testing Images	2,520
Number of Classes	15

VII. PROBLEM STATEMENT

Recognizing human actions from images is an important computer vision task. Variations in body posture, camera angles, lighting conditions, and background complexity make action recognition challenging. The objective of this project is to develop a CNN-based framework capable of automatically recognizing human actions from image data.

VIII. RESEARCH QUESTIONS

This study aims to answer the following research questions:

- RQ1: Can a CNN effectively recognize human actions from images?
- RQ2: What classification accuracy can be achieved on the action recognition dataset?
- RQ3: Which actions are most difficult to classify?
- RQ4: How does CNN performance improve during training?

IX. DATASET PREPROCESSING

Before training, all images were resized to 224×224 pixels to ensure a consistent input size. Pixel values were normalized to improve numerical stability during training. The activity labels were encoded into numerical categories for multi-class classification.

X. METHODOLOGY

A Convolutional Neural Network (CNN) was developed to classify human activities from images. The methodology consisted of the following stages:

- 1) Dataset Loading
- 2) Data Preprocessing
- 3) Image Resizing
- 4) Data Normalization
- 5) CNN Model Construction
- 6) Model Training
- 7) Model Evaluation

Images were resized to 224×224 pixels before training. Pixel values were normalized to improve model convergence. The CNN automatically extracts hierarchical visual features and performs multi-class classification through a Softmax output layer.

XI. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was performed to understand the dataset structure and distribution of classes.

The dataset contains images representing different human activities captured under varying lighting conditions, camera angles, and backgrounds. Data exploration revealed that the dataset is well organized with predefined training and testing subsets.

Image preprocessing included resizing images to a fixed resolution and normalizing pixel values before training.

XII. MODEL ARCHITECTURE

The CNN model consists of three convolutional layers followed by max pooling operations for feature extraction. The extracted features are flattened and passed through a dense layer with dropout regularization before the final classification layer.

The model contains 22,248,527 trainable parameters.

The CNN architecture consists of three convolutional layers, three max-pooling layers, a flatten layer, a dense hidden layer, dropout regularization, and a final softmax classification layer with 15 output classes.

Total Parameters: 22,248,527

Trainable Parameters: 22,248,527

Non-Trainable Parameters: 0

A Convolutional Neural Network (CNN) architecture was used for image classification.

The model consists of:

- Convolution Layers
- ReLU Activation Functions
- Max Pooling Layers
- Dropout Layers

TABLE II
CNN MODEL SUMMARY

Layer (Type)	Output Shape	Parameters
Conv2D	(None, 222, 222, 32)	896
MaxPooling2D	(None, 111, 111, 32)	0
Conv2D	(None, 109, 109, 64)	18,496
MaxPooling2D	(None, 54, 54, 64)	0
Conv2D	(None, 52, 52, 128)	73,856
MaxPooling2D	(None, 26, 26, 128)	0
Flatten	(None, 86528)	0
Dense	(None, 256)	22,151,424
Dropout	(None, 256)	0
Dense	(None, 15)	3,855

- Fully Connected Dense Layers
- Softmax Output Layer

The CNN automatically learns discriminative visual features from human activity images and maps them to their corresponding action classes.

XIII. EXPERIMENTAL SETUP

Experiments were conducted using TensorFlow and Keras in the Kaggle notebook environment.

TABLE III
TRAINING PARAMETERS

Parameter	Value
Image Size	224 × 224
Batch Size	32
Epochs	10
Optimizer	Adam
Loss Function	Categorical Crossentropy
Output Classes	15

The CNN model contained approximately 22.25 million trainable parameters.

XIV. RESULTS AND DISCUSSION

The CNN model was trained on the Human Action Recognition dataset containing 15 activity classes.

During the first training epoch, the model achieved an accuracy of 9.20% and a loss value of 3.0105. Since the model was at the initial stage of learning, the classification accuracy was relatively low. Additional epochs are expected to improve performance as the network learns more discriminative features from the training images.

The model requires additional epochs to achieve higher classification performance. The reported accuracy corresponds to the initial stage of training and serves as a baseline for further optimization.

TABLE IV
INITIAL TRAINING PERFORMANCE

Metric	Value
Epoch	1/10
Training Accuracy	9.20%
Training Loss	3.0105
Training Steps Completed	166/315

Table IV presents the initial training performance.

XV. FUTURE WORK

Future work may involve implementing transfer learning using pretrained architectures such as ResNet50, MobileNetV2, and EfficientNet. Additional improvements may include data augmentation, hyperparameter tuning, and evaluation on larger action recognition datasets.

XVI. CONCLUSION

This project developed a deep learning-based Sports Action Recognition Framework using the Human Action Recognition Dataset. A CNN model was trained to recognize and classify human actions from images.

The study demonstrates that deep learning techniques can effectively solve multi-class image classification problems. Future work may investigate transfer learning using architectures such as ResNet50, MobileNetV2, and EfficientNet to further improve performance.

ACKNOWLEDGMENT

The author would like to thank the University of Europe for Applied Sciences, the course instructors, and the Kaggle community for providing the dataset and computational resources used in this study.

REFERENCES

- [1] Y. LeCun, Y. Bengio and G. Hinton, "Deep Learning," Nature, 2015.
- [2] A. Krizhevsky, I. Sutskever and G. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," NIPS, 2012.
- [3] K. He, X. Zhang, S. Ren and J. Sun, "Deep Residual Learning for Image Recognition," CVPR, 2016.
- [4] Meet Nagadia, "Human Action Recognition Dataset," Kaggle, 2025.
- [5] K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," International Conference on Learning Representations (ICLR), 2015.
- [6] A. G. Howard et al., "MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications," arXiv:1704.04861, 2017.
- [7] M. Tan and Q. V. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," Proceedings of ICML, 2019.
- [8] J. Wang, Y. Chen, S. Hao, X. Peng, and L. Hu, "Deep Learning for Sensor-Based Activity Recognition: A Survey," Pattern Recognition Letters, vol. 119, pp. 3–11, 2019.
- [9] K. Soomro, A. R. Zamir, and M. Shah, "UCF101: A Dataset of 101 Human Actions Classes From Videos in the Wild," CRCV-TR-12-01, University of Central Florida, 2012.
- [10] J. Deng et al., "ImageNet: A Large-Scale Hierarchical Image Database," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2009.

- [11] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A Simple Way to Prevent Neural Networks from Overfitting," *Journal of Machine Learning Research*, vol. 15, pp. 1929–1958, 2014.
- [12] M. Abadi et al., "TensorFlow: Large-Scale Machine Learning on Heterogeneous Systems," Google Research, 2015.
- [13] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," *International Conference on Learning Representations (ICLR)*, 2015.
- [14] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-Based Learning Applied to Document Recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.